Lotka - Volterra

Drone simulation is fundamentally a dynamic system: Its state changes over time according to differential equations.

Lotka-Volterra is a classic non-linear dynamic system that models interaction between two populations - a prey and a predator.

What is Lotka-Volterra :-

Lotka-Volterra predator-prey model describes two interacting species.

x(t) : prey population

y(t) : predator population

$dx/dt = \alpha x - \beta xy$	αx : prey grows naturally when food is available
	$-\beta xy$: prey decreases due to encounter with predators
$dy/dt = -ry + \delta xy.$	$-ry$: predator die naturally without food
	$+\delta xy$: predators increase when they encounter prey.

The key idea is "Coupling" : prey affects predators and predators affect prey. This is similar to how a drone's roll angle affects velocity, which affects position, which affects the controller's correction.

Intuition behind the oscillation :-

1. prey population increases
2. More prey means predator get more food, so predators increases
3. More predators reduces the prey population
4. With less prey, predator begin to starve and decrease.
5. With fewer predators prey recovers.

So the system naturally produces delayed oscillations : prey rises first then predators rise, then prey falls, then predator falls.

Equilibrium points :-

Equilibrium ⇒ system stops changing

$dx/dt = 0$ $dy/dt = 0$

In the standard model,

$x(\alpha - \beta y) = 0$
$-y(r - \delta x) = 0$ } This gives two equilibria (0,0) and $[r/\delta , \alpha/\beta]$

(0,0) indicates both species are extinct.

$(r/\delta , \alpha/\beta)$ coexistance equilibrium, where births and deaths balance.

example:

$\alpha = 1$ $\beta = 0.1$ $r = 1.5$ $\delta = 0.0075$

$x^* = r/\delta = 1.5 / 0.075 = 20$ $y^* = \alpha/\beta = 1 / 0.1 = 10$

So the coexistance equilibrium is (20,10)

At this point, prey and predator population remains constant in the ideal model.

Toy numerical example :

$\alpha = 1 \quad \beta = 0.1 \quad r = 1.5 \quad \delta = 0.075$
$x(0) = 40 \quad y(0) = 5 \rightarrow$ Initial populations.

using Euler's method,

$x_{n+1} = x_n + h(\alpha x_n - \beta x_n y_n)$
$y_{n+1} = y_n + h(-r y_n + \delta x_n y_n)$

Euler's method : Approximating the solution to a first-order ordinary differential equation, given an initial condition.

Take h = 0.1

At t = 0,

$x_0 = 40 , \quad y_0 = 5$

$dx/dt = 1(40) - 0.1(40)(5) = 40 - 20 = 20$
$dy/dt = -1.5(5) + 0.075(40)(5) = -7.5 + 15 = 7.5$

$x_1 = 40 + 0.1(20) = 42$
$y_1 = 5 + 0.1(7.5) = 5.75$

At this moment prey is increasing, and because there is plenty of …

Coexistance equilibrium of the basic Lotka-Volterra model is a center ; nearby trajectories orbit around it rather than converging to it. That's why model is giving sustained oscillations.

Jacobian matrix at equilibrium,

$$J = \begin{vmatrix} \alpha - \beta y & -\beta x \\ \delta y & -r + \delta x \end{vmatrix}$$

At the coexistance equilibrium , $x^* = r/\delta$ and $y^* = \alpha/\beta$

Eigen value, $\lambda = \pm i\sqrt{(\alpha r)}$: purely imaginary.

Neutral stability : The system oscillates indefinitely, amplitude depending on initial condition

In Drone, we usually want stable equilibrium - hover at a desired altitude or position. In basic-Lotka Volterra, the equilibrium is neutrally stable, so perturbations keep running instead of running out. More realistic models add damping, carrying capacity or control-like terms to make the system converge.

Stability determined through eigen value of the Jacobian

Positive real part ⇒ Instability.
Negative real part ⇒ Local asymptotic stability

Relevance in Drone simulation :-

* Same core idea

- Model the system using differential equations
- Identify the state variables
- Find the equilibra
- Analyse stability using Jacobian and eigen values.
- Simulate non-linear dynamics numerically.

In drones states are, position, velocity, altitude and angular rates and in Lotka Volterra it is prey and predator population. The mathematic part is the same.

For Lotka-Volterra

* Equilibrium is stable in the sense, the nearby states do not fly to infinity
* But it is not asymptotically stable, because the state doesn't approach the equilibrium over time.
* This is neutral stability.

In drones, PID controller brings altitude error to zero; drone settles at a target altitude.